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AS THE INDIAN Space Research Organisation (Isro) gears up for Gaganyaan, its first human spaceflight mission, scheduled for 2025, the excitement is palpable. The question is not whether Isro can send a crew into a low-Earth orbit but when a woman will get there. (The four candidates training for the first flight are all men and Indian Air Force pilots.) For Isro, sending a woman into space is not a big deal; it's the next logical step. Women account for over a quarter of its scientific and technical workforce, holding key posts, leading teams and driving innovation.

Take Neelavathy M., 58, senior scientist and Deputy Director at Isro's UR Rao Satellite Centre, Bengaluru. Her first significant achievement was nearly a decade ago when she led the team that designed a communication system for Mangalyaan, the Mars Orbiter Mission or MOM.

She also recalls her recent responsibility of leading a team configuring the GSAT-7 series of satellites. "With tight deadlines and immense pressure, we worked in perfect sync," Neelavathy says. "We had no time to think in gender terms. The focus was on solving the problem."

The camaraderie she experienced in that project epitomises Isro's culture: success is a collective triumph. When the applause erupted first for the successful launch of the Mars orbiter on November 5, 2013, and then for the orbiter's successful insertion in a Mars orbit on September 24, 2014, and the Chandrayaan-3 Moon landing, it was a victory shared by Isro's men and women. "At Isro, gender has never dictated who gets the top assignments," Neelavathy says. "Your technical capability and ability to inspire confidence in your team members determine your success."

In the 38 years since Neelavathy joined as a Scientist/Engineer-SB, her career has seen many milestones, such as designing and delivering telemetry and ranging systems for over 30 satellites, including Chandrayaan-1 and Chandrayaan-2.

Neelavathy's launchpad was a lecture on satellite communications

in her engineering studies class at Thiagarajar College in Madurai. "I spent hours in the library reading up the subject and got into satellite communications," she recalls. Today, she oversees the development and management of spacecraft power, navigation, and communication systems.

Neelavathy is among the many women in leadership roles at Isro. Women have played crucial roles in Chandrayaan-1 and Chandrayaan-2, India's lunar missions, from mission planning to scientific analysis. Women engineers and scientists had a role in the success of the MOM, which made India the first Asian nation to reach Mars' orbit and Isro the world's fourth agency to do so.

CATALYST FOR CHANGE

Over a quarter of Isro's workforce of 16,000 are women (20-25% or 3,200-4,000). The Chandrayaan-3, which landed near the Moon's south pole on August 23, 2023, involved 100-plus women scientists and engineers.

At the India Space Congress 2024, an annual networking event organised by the SatCom Industry Association, women's participation increased by 15% over the previous year. India's space sector is attract-

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SOARING HIGH



► The Indian space sector is witnessing a rise in female participation, particularly in the STEM fields, thanks to government initiatives like 'Women in Science'



► Women currently make up nearly 30% of the workforce in India's space sector, including roles in Isro and private space companies, up from just 12% in 2019



► A push towards privatisation and commercialisation is also creating opportunities for women in the private space sector as a lot of start-ups are entering the segment



► Start-ups like SpaceKidz, which designed and launched satellites with the help of 750 girls, also inspire young girls to pursue careers in the space sector